

The Effect of Punishment on Firm Compliance with Hazardous Waste Regulations¹

Sarah L. Stafford

*Department of Economics, College of William & Mary, Morton Hall,
Williamsburg, Virginia 23187-8795
E-mail: slstaf@wm.edu*

Received June 22, 2000; revised February 6, 2001; published online January 11, 2002

This paper examines the effect of EPA's dramatic 1991 increase in the penalties associated with violations of hazardous waste regulations on firm compliance. Although data from EPA's RCRA Information System show an increase in detected violations after the penalty change, once inspection levels are incorporated into the analysis through a censored bivariate probit model I find that violations have decreased since the penalty change. However, the decrease in violations appears small relative to the increase in recommended penalty levels. I also find evidence that the likelihoods of inspection and compliance vary significantly across regions.

© 2002 Elsevier Science (USA)

In 1990 the U.S. Environmental Protection Agency (EPA) published the Revised RCRA Civil Penalty Policy. This policy significantly increased the penalties associated with violations of hazardous waste regulations issued under the Resource Conservation and Recovery Act (RCRA). This revision was issued in response to two reports that criticized EPA for setting low and inconsistent penalties for hazardous waste violations.² Under the new policy, some penalties were increased to 10 or 20 times the previous penalty levels. In addition, penalties were structured to "fit the crime" as the size of the penalty depends on the probability of harm and the extent of deviation from the regulatory requirements. Penalties also can be increased if the facility has a record of non-compliance. According to William Frank, special assistant in EPA's Office of Enforcement at the time, "In the past, some companies considered RCRA fines to be the cost of doing business, but they won't anymore."³

This paper examines whether the revised penalty policy had the intended effect of increasing compliance with hazardous waste regulations. The remainder of this paper is organized as follows. Section 1 provides a discussion of the related literature and Section 2 provides a brief background on hazardous waste regulation and enforcement. Section 3 describes the empirical approach and Section 4 discusses the data used in the analysis. Section 5 provides the results of the analysis and Section 6 presents some conclusions.

¹ I am grateful to Jennifer Mellor, Seth Carpenter, Arik Levinson, participants in the March 1999 CCOFFE Workshop, and three anonymous referees for their comments.

² "U.S. EPA Ups the Ante: Companies Facing New Penalties in for Sticker Shock," *EI Digest*, February 1991.

³ "EPA Raises the Price of Delay on RCRA Compliance," *Chemical Week*, November 7, 1990.

1. RELATED LITERATURE

Becker [1] provides the seminal paper on the economics of crime and enforcement. Assuming rational behavior, a crime will be committed if the expected value of the crime is greater than the expected cost of the crime. The expected cost depends in part on the penalty associated with the crime, and compliance can always be enhanced (at least weakly) by increasing penalties. However, other models have shown that increasing penalties may not increase compliance. Leung [12] shows that in a dynamic model of crime, as opposed to Becker's static model, maximal punishments are not necessarily optimal. Similarly, Bose [2] shows that in a model with regulatory errors, the optimal penalty is non-maximal. In this model, higher penalties can lead to efforts to obstruct the enforcement process such as increasing the incentives to challenge the penalty in court. With respect to environmental regulations, the literature exhibits the same contradictions. For example, Heyes [8] shows that penalties should be lowered if imposing them discourages cleanup while Lear and Maxwell [11] show that maximum penalties can be optimal if a firm's ability to pay is taken into account.

The existing empirical literature on compliance with environmental regulations has focused primarily on the relationship between compliance and enforcement. Magat and Viscusi [13] examine plant compliance with water pollution regulations in the U.S. pulp and paper industry during 1982 to 1985. The authors model compliance as a function of past pollution and inspections, plant capacity, industry, and region and find that an increase in inspections, their proxy for enforcement efforts, increases firm compliance with regulations.⁴ Laplante and Rilstone [10] also examine the compliance of firms in the pulp and paper industry with water pollution regulations although they look at Canadian firms. In addition, Laplante and Rilstone estimate the effect of compliance on inspections while controlling for the endogeneity of inspections. They find that both past inspections and anticipated future inspections increase compliance.⁵

Gray and Deily [3] analyze compliance with and enforcement of air pollution regulations in U.S. steel plants. First, the authors estimate compliance as a function of plant and local characteristics and enforcement as a function of plant, local, and state characteristics. These predicted levels of enforcement and compliance are then used in the compliance and enforcement equations, respectively. The authors find that higher levels of enforcement encourage higher levels of compliance but that higher levels of compliance result in lower levels of enforcement. Helland [7] analyzes the U.S. pulp and paper industry with respect to water pollution regulations. He uses a detection-controlled estimation model to control for missing data and estimates both compliance and inspections. In contrast to earlier studies, this study finds that compliance rates are not affected by inspections. However, noncompliance appears to encourage inspections.

A few empirical studies have included penalties explicitly in their analysis of environmental enforcement and compliance. For example, using data on water

⁴ Magat and Viscusi reject the hypothesis that contemporaneous inspections are exogenous. However, because using instrumental variables for contemporaneous inspections leads to positive coefficients, the authors believe that using the lagged inspection variable does not create a bias.

⁵ Because the information on pollution is self-reported to some extent, Laplante and Rilstone also estimate a reporting equation to correct for possible sample selection bias.

pollution violations in Louisiana, Kleit *et al.* [9] analyze the relationship between the severity of environmental violations and the penalties facilities face. The authors find that penalties are more likely to be assessed and are higher for more severe violations. Additionally, past violations increase the likelihood and severity of penalties. Hamilton [6] examines the effect of informal and formal rules in enforcing RCRA regulations. He finds that environmental penalties increase with the severity of environmental violations and the cost of compliance. Hamilton also finds that penalties are higher in locations with legislators on RCRA oversight committees or with a high level of membership in environmental groups.

None of the existing empirical literature on compliance with environmental regulations has explored the effect that a change in expected penalty level has on firm compliance. Although some studies have included penalties, for the most part penalties have been considered as a dependent, rather than explanatory, variable. This study, on the other hand, examines the effect that an increase in expected penalties has on compliance behavior. According to the theoretical models, it is the expected penalty (rather than the actual penalty) that should affect compliance behavior. Additionally, very little of the existing literature on compliance behavior has focused on hazardous waste regulations. This paper uses a censored bivariate probit model to examine the effect of an increase in penalty levels on compliance with hazardous regulation across a wide variety of industries in the United States.

2. BACKGROUND ON HAZARDOUS WASTE REGULATION AND ENFORCEMENT

Under Subtitle C of RCRA, EPA regulates hazardous waste from “cradle to grave,” that is, from generation to ultimate disposal. Under RCRA, EPA has established standards with which all facilities that generate, store, transport, treat, or dispose hazardous waste must comply.⁶ RCRA also provided EPA with the ability to enforce the act by allowing it to assess civil penalties upon firms that violate RCRA requirements.

A hazardous waste generator is the legal entity, usually the owner or operator of an industrial process, that produces a hazardous waste or first causes a waste to become subject to RCRA regulation. A generator is responsible for determining if a solid waste is hazardous and, if it is, must notify EPA that the facility generates hazardous waste and ensure that the waste is managed in accordance with RCRA standards. Facilities that treat, store, or dispose hazardous waste also must comply with RCRA management standards. Some of these facilities must obtain a RCRA operating permit that establishes the site-specific standards with which the facility must comply (in addition to the generic management standards). Hazardous waste may be managed either at the site of generation or off site at another facility. If the waste is managed off site, it must be packaged according to RCRA standards and transported by a firm licensed to transport hazardous waste.

Violations of RCRA regulations may be willful or inadvertent. A facility may knowingly violate the regulations, for example by sending hazardous waste to a

⁶ Environmental damages from hazardous waste at inactive or abandoned sites (as opposed to ongoing operations) as well as hazardous waste resulting from spills that require an emergency response are covered by the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) of 1980, commonly referred to as Superfund, rather than by RCRA.

nonhazardous waste landfill for disposal. A facility also may unknowingly violate the regulations if one of its storage tanks is leaking, for example. In the case of inadvertent violations, plant operators and owners have choices over types of equipment that can affect the probability that a violation will occur. EPA enforces RCRA regulations primarily through facility inspections, although violations that result in damage to human health or the environment are sometimes self-reported or reported by third parties. All facilities are required to undergo a compliance evaluation inspection once every two years, although facilities may be inspected more often.

As with all environmental regulations, EPA has established the minimum level (the federal level) of hazardous waste regulations. States can make additional regulations concerning hazardous waste management that are more stringent than the federal regulations. EPA will authorize a state to run its own hazardous waste program as long as the state's program is at least as stringent as the federal program. The authorized states implement and enforce their own hazardous waste regulations, and EPA, through its regional offices, implements and enforces the regulations in the nonauthorized states.

Once a violation is discovered there is a range of enforcement options including administrative, civil, and criminal actions that can be used, depending on the severity of the violation. All types of actions can include penalty provisions. Under the revised penalty policy, penalties must reflect the seriousness of the violation but can be adjusted based on a facility's past compliance history. Although those setting the penalties are not required to follow the penalty policy, they must consider it and explain any deviations from it.

3. EMPIRICAL APPROACH

The key question this paper seeks to answer is whether EPA's dramatic increase in the penalties associated with violations of hazardous waste regulations in 1991 affected the compliance behavior of firms. As Table I shows, the number of detected violations increased after 1991, although the number has been declining since 1992. Taken at face value, one might conclude that compliance decreased after the 1991 penalty increase. However, in terms of the number of detected violations per inspection, one can see a trend of increasing violations relative to inspections from 1986 to 1992, with a decreasing trend from 1992 onward. To determine how firm compliance has actually changed in response to the 1991 penalty increase, one must account for the increase in inspection rates as well as any changes in inspection targeting that could affect the rate at which violations are detected.

Figure 1 illustrates the firm's compliance decision. If the firm complies with the regulation, regardless of whether it is inspected, it receives no fine. Of course, this assumes no false positive results or at least that proving compliance is costless.⁷ If the firm violates the regulation, with probability s it will not be inspected and thus will receive no fine. However, with probability $(1 - s)$ it will be inspected and will

⁷ If it is not costless to prove compliance, or if inspections are costly, this can be incorporated into the model of the firm's decision. In such a case, if firms can decrease inspection rates by compliance, firms may comply even if the cost of compliance is higher than the expected penalty.

TABLE I
Annual RCRA Inspections and Violations, 1986–1995

Year	Number of inspections ^a	Number of violations ^b	Violations per inspection
1986	20,855	6,614	0.32
1987	21,401	7,585	0.25
1988	22,981	8,702	0.38
1989	23,109	9,881	0.43
1990	24,425	11,121	0.46
1991	27,292	16,128	0.59
1992	28,286	18,174	0.64
1993	29,354	17,974	0.61
1994	29,997	17,005	0.57
1995	28,889	14,127	0.49

Source: RCRIS, July 1999.
^a Total number of Inspections reported.
^b Total number of Class 1 Violations reported.

be assessed the fine F .⁸ Therefore, if the cost of compliance is C , firms should comply when $C < (1 - s)F$.

For simplicity, the model assumes that all violations are willful, that is there are no inadvertent violations.⁹ The cost of compliance, C , is obviously firm specific as it will depend on the type and quantity of waste the firm generates. The probability of inspection, s , will also be firm specific as EPA targets sites based on human health and environmental risks and past compliance history. Finally, the penalty if the firm is caught, F , is firm specific, as the revised RCRA penalty policy takes the gravity of the violation, the economic benefit gained through noncompliance, and the site compliance history into account.

⁸ This assumes that detection is complete. A more complex model would allow the probability of detection be less than 1, and perhaps endogenous. Additionally, the model assumes that there are no false positives, that is, that EPA cannot find a violation where none exists.

⁹ Inadvertent violations will not affect the model as long as they are white noise. If firms are less likely to be fined if a violation is accidental, then the results of this analysis will be biased towards 0.

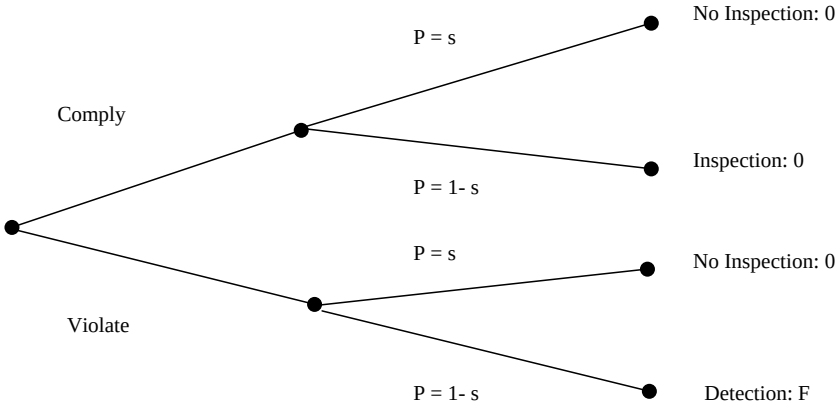


FIG. 1. Firm's decision tree.

Most of the existing empirical work on environmental compliance has used data sets with monthly observations of whether the firm is in compliance.¹⁰ In contrast, the data set for this analysis (discussed in detail in Section 4) contains information on compliance only if the facility is inspected. If the facility is not inspected there is no information about whether the facility is in compliance. Because the data are censored, I construct a censored bivariate probit similar to the type used in Greene [4].

The probability of a violation is modeled as a latent variable Y_{1i}^* that is a function of explanatory variables. Y_{1i} is the potentially observable binary variable equal to 1 if the facility violates the regulations

$$Y_{1i}^* = x_{1i} \beta_1 + \varepsilon_{1i}$$

$$Y_{1i} = \begin{cases} 1 & \text{if } Y_{1i}^* > 0 \\ 0 & \text{if } Y_{1i}^* \leq 0. \end{cases}$$

The probability of an inspection is also modeled as a latent variable, Y_{2i}^* . Y_{2i} is equal to 1 if the facility is inspected:

$$Y_{2i}^* = x_{2i} \beta_2 + \varepsilon_{2i}$$

$$Y_{2i} = \begin{cases} 1 & \text{if } Y_{2i}^* > 0 \\ 0 & \text{if } Y_{2i}^* \leq 0. \end{cases}$$

EPA is more likely to inspect facilities that it suspects of being in violation. Therefore, if there are unmeasured characteristics of the facility that make it more likely to violate and thus more likely to be inspected, the error terms ε_{1i} and ε_{2i} should be positively correlated. Thus, the likelihood of observing a detected violation ($Y_{1i} = Y_{2i} = 1$) is modeled as

$$L_{Y_{1i}=1, Y_{2i}=1} = \sum_{Y_{1i}=1, Y_{2i}=1} \log\{\Phi_2[x_1 \beta_1, x_2 \beta_2, \rho]\},$$

where Φ_2 is the bivariate normal cumulative distribution function and ρ is the covariance between ε_{1i} and ε_{2i} .

The likelihood of inspecting a non-violator ($Y_{1i} = 0, Y_{2i} = 1$) is

$$L_{Y_{1i}=0, Y_{2i}=1} = \sum_{Y_{1i}=0, Y_{2i}=1} \log\{\Phi_2[-x_1 \beta_1, x_2 \beta_2, -\rho]\}.$$

Because Y_{1i} is observed if and only if Y_{2i} is equal to 1, a nonviolating facility that is not inspected is observationally equivalent to a violating facility that is not inspected. Therefore the maximum likelihood function for the censored bivariate probit can be expressed as

$$L = L_{Y_{1i}=0, Y_{2i}=1} + L_{Y_{1i}=1, Y_{2i}=1} + L_{Y_{2i}=0}$$

$$= \sum_{Y_{1i}=1, Y_{2i}=1} \log\{\Phi_2[x_1 \beta_1, x_2 \beta_2, \rho]\}$$

$$+ \sum_{Y_{1i}=0, Y_{2i}=1} \log\{\Phi_2[-x_1 \beta_1, x_2 \beta_2, -\rho]\} + \sum_{Y_{2i}=0} \log\{1 - \Phi[x_2 \beta_2]\}.$$

¹⁰ Most of this data is self-reported by the facilities.

4. DESCRIPTION OF DATA

The primary data used for this study come from EPA's RCRA Information System (RCRIS). RCRIS contains information on inspections and violations at hazardous waste generation and management facilities from 1980 to the present.¹¹ Although RCRIS contains data from 1980, the coverage for earlier years is not complete. Also, EPA's regulations have changed significantly since 1980. Therefore, this study begins in 1986, after the passage of the Hazardous and Solid Waste Amendments (HSWA) to RCRA that implemented the core regulations on hazardous waste facilities. The last year analyzed is 1995, because beginning in 1996 EPA implemented a self-audit policy that reduces the penalty if the facility reports violations that it discovers during a self-audit.

Not all of the facilities in RCRIS are included in the analysis. To determine how the change in penalty structure affected compliance behavior, I wanted to analyze the compliance behavior of facilities that were subject to RCRA both before and after the 1991 penalty increase. Therefore I constructed a set of 8411 facilities that have been generating and/or managing hazardous waste consistently over the period of the analysis.¹² This group excludes federal facilities because federal facilities were not subject to RCRA penalties until after passage of the Federal Facilities Compliance Act in 1992. It also excludes small quantity generators because the regulations governing small quantity generators are significantly more lenient than those governing large quantity generators.¹³ The set of facilities used in the analysis represents approximately 40% of the 1995 universe of large quantity generators and treatment, storage, and disposal facilities and accounts for almost 90% of hazardous waste managed in 1995.

As discussed in Section 3, the censored bivariate probit consists of two equations, the Inspection equation and the Violation equation. For the Inspection equation, the dependent variable is a dummy variable equal to 1 if the facility is inspected in year t .¹⁴ For the Violation equation, the dependent variable is a dummy variable equal to 1 if any RCRA violation is discovered at the facility in year t .¹⁵ Whether a facility is inspected depends on the characteristics of the facility as well as the state's enforcement program. Whether a facility will find it optimal to violate the environmental regulations depends on the benefits from violation (in this case, avoided compliance costs), the probability that the facility will be inspected and thus the violation detected, and the penalty associated with the violation. Table II presents the variables used in the Inspection and Violation

¹¹ The data in RCRIS are at the facility, or plant, level rather than at the firm level.

¹² To construct this set of facilities, I used information from EPA's Biennial Reporting System for 1989, 1991, 1993, and 1995 to determine the facilities that had either generated or managed some hazardous waste in each of these four survey years. The time trends in terms of violations and inspections for this smaller set of facilities are very similar to the time trends reported in Table I for all RCRA facilities.

¹³ A facility is a small quantity generator if it does not generate more than 1000 kg of hazardous waste in any single month.

¹⁴ In constructing this variable, all types of inspections recorded in RCRIS were counted.

¹⁵ I also considered a second specification for the violation equation where the dependent variable is a dummy variable equal to 1 if a Class 1 violation is discovered at the facility in year t . Class 1 violations are a subset of RCRA violations and are more serious than other violations. The results of the analysis when Class 1 violations are used as the dependent variable do not differ qualitatively from the results when all violations are used and are therefore not reported in this paper.

TABLE II
Variables Used in the Inspection and Violation Equations
(Annual Data 1986–1995 for 8411 Facilities)

Variable	Definition	Exp. sign in Inspection eq.	Exp. sign in Violation eq.	Mean
Inspection	Dummy = 1 if inspected.	NA	NA	0.39
Violation	Dummy = 1 if violation discovered.	NA	NA	0.24
Permit	Dummy = 1 if facility has RCRA permit.	+	?	0.10
Generated	Log of tons of waste generated.	+	?	5.03
Managed	Log of tons of waste managed.	+	?	−4.80
Received	Log of tons of waste received for management.	+	?	−10.66
High Priority	Dummy = 1 if facility is high priority for corrective action.	+	−	0.10
Medium Priority	Dummy = 1 if facility is medium priority for corrective action.	+	−	0.08
Inspected Last	Dummy = 1 if inspected last year.	−	+	0.39
Violated Last	Dummy = 1 if violation found last year.	+	−	0.24
TC	Dummy = 1 if facility generated or managed TC waste.	?	?	0.14
State Inspections	Number (in hundreds) of inspections in state, normalized by gross state product.	+	−	0.50
State Violations Last	Number (in hundreds) of violations in state last year, normalized by gross state product.	−	NA	0.20
Strict Liability	Dummy = 1 if state had strict liability regime.	+	−	0.78
Gross Product	Gross state product (in million dollars).	−	+	0.23
Environmental Membership	State membership in environmental organizations, out of 1,000.	NA	−	8.61
Post-Change	Dummy = 1 if post-revised penalty (1992–1995).	+	?	NA
Transition	Dummy = 1 if transition year (1990–1991).	+	?	NA

equations, the expected sign of the coefficient in the relevant equations, and the means of these variables. Unless otherwise noted, the data described are extracted from RCRIS.

The first set of explanatory variables describes the nature and level of hazardous waste activity at the facility. Most facilities that treat or dispose of hazardous waste on site must obtain a RCRA operating permit and are subject to more onerous regulations than facilities that do not manage hazardous waste on site. Also, one could argue that there is more potential for serious environmental damages at facilities that manage hazardous waste than at facilities that only generate hazardous waste, and thus enforcement authorities are likely to target such facilities. Therefore the dummy variable Permit is included in the model.¹⁶ However, it is not obvious whether the coefficient on Permit in the Violation equation will be positive or negative. On one hand, permitted facilities must comply with a larger set of

¹⁶ Including additional variables to describe more specifically the nature of waste management at a facility such as a dummy variable indicating whether the facility had a hazardous waste combustion unit did not significantly improve the results of the analysis.

regulations so the costs of compliance are higher for these facilities, suggesting that permitted facilities have stronger incentives to violate the regulations. On the other hand, because permitted facilities are more likely to be targeted by enforcement authorities, they may be more likely to comply than nonpermitted facilities.¹⁷

The next three variables, Generated, Managed, and Received, measure the level of various hazardous waste activities at the facility in year t and are derived from EPA's Biennial Reporting System.¹⁸ In general, higher levels of activity increase compliance costs as well as the potential for damages. All three variables are included because the impact of each type of activity on compliance costs could be different. For example, facilities that generate hazardous waste but do not manage the waste on site have significantly lower compliance costs than facilities that manage the same quantity of waste on site. Also, facilities that receive waste from off-site for management, i.e., commercial hazardous waste facilities, may be inspected more frequently than noncommercial facilities. In each case, the effect of these variables on the Violation equation is ambiguous because although there should be an increased likelihood of inspection, higher levels of activity will also increase the facility's cost of compliance, and thus the potential benefit from violating the regulation.

High Priority is a dummy variable equal to 1 if the facility is considered a high-priority facility in the RCRA corrective action program and Medium Priority is a dummy variable equal to 1 if the facility is considered a medium-priority facility.¹⁹ Facilities that are considered high or medium priority for corrective action have had releases to the environment in the past, and thus are more likely to be targeted for inspection. Therefore, the coefficients on High Priority and Medium Priority are both expected to be positive in the Inspection equation and negative in the Violation equation.

Inspected Last is a dummy variable equal to 1 if the facility was inspected in year $t - 1$. Inspected Last is included in the Inspection equation because, according to RCRA policy, every facility should be inspected at least once every two years. If the facility was inspected in the previous year, it is less likely to be inspected this year, *ceteris paribus*. In the Violation equation, I expect the coefficient on Inspected Last to be positive, as it will decrease the facility's expected probability of detection.

Violated Last is a dummy variable equal to 1 if the facility was discovered violating any RCRA regulations in year $t - 1$. The coefficient on this variable in the Inspection equation should be positive, as EPA will conduct follow-up inspections to ensure that past offenders have come into compliance with the regulations.

¹⁷ Also, permitted facilities may be more concerned with their records than nonpermitted facilities, as violations could hurt the facility's chances of getting a permit renewal.

¹⁸ The Biennial Reporting System (BRS) is a census of hazardous waste generation and management facilities conducted by EPA in odd years. The BRS contains information on the amount of waste generated by the facility, the amount of waste received by the facility from other generators, and the total amount of waste managed on-site. Because the survey is conducted biennially, the quantities for the even years are interpolated. Also, 1989 was the first year with complete data. Therefore for years prior to 1989, the quantities used in the analysis are the 1989 quantities. Excluding these three variables from the model does not significantly change the results of the analysis.

¹⁹ The RCRA corrective action program oversees the cleanup of nonactive waste management units at active waste management facilities. Therefore the violations considered in this analysis, which occur at active waste management units, should not affect the prioritization of facilities for the corrective action program.

Because this will increase the expected probability of detection at a facility and because penalties can be increased if the facility has a past history of violations (at least after the change in the penalty policy), I expect the coefficient on Violated Last to be negative in the Violation equation.

TC is a dummy variable equal to 1 if the facility generated or managed any toxicity characteristic (TC) waste in year t .²⁰ In 1990, EPA passed the TC Rule, which significantly increased the universe of hazardous waste regulated under RCRA. Although passage of the TC Rule roughly coincided with the revision of the RCRA penalty policy, not all facilities in the analysis were affected by the TC Rule. Therefore this variable should capture the effects of the TC Rule on inspections and compliance behavior.

The next set of explanatory variables captures various state characteristics. Because states vary in enforcement programs, both the likelihood of inspections and the expected penalty for violations should vary across states. The first variable, State Inspections, measures the number of inspections (in hundreds) conducted in the state, normalized by the gross product of the state.²¹ Because a higher level of inspection activity increases the probability that a given facility will be inspected, the coefficient on this variable should be positive in the Inspection equation. Additionally, the number of inspections should be directly correlated with the amount of resources devoted to overall hazardous waste program implementation and enforcement. One might expect penalties to be higher in states that spend more resources on hazardous waste implementation and enforcement, in which case the coefficient on State Inspections in the Violation equation should be negative.

State Violations Last measures the number of violations detected in the state in year $t - 1$, normalized by the gross product of the state.²² This variable is included because if there were many violations in the last year, the state might have to devote more resources to conducting follow-up inspections this year, and thus fewer general inspections will be made at any one facility. As discussed in Poirier [14], the bivariate probit model will be identified if there is at least one variable that affects whether the facility will be inspected but does not affect whether the facility violates the regulations, as well as at least one variable that affects whether the facility violates the regulations but does not affect whether the facility will be inspected. State Violations Last is included in the Inspection equation but is not included in the Violation equation. Although the number of violations in the state last year should affect the probability that any inspection will be conducted, data

²⁰ Using the BRS, I identified all facilities that generated or managed any waste stream that was hazardous only due to the TC constituents. Because the TC Rule did not take effect until 1991, TC is equal to 0 for all facilities prior to 1991. Beginning in 1991, TC is equal to 1 for approximately 30% of the facilities in the analysis. Note that since the analysis only includes facilities that were active during the entire period of analysis, there are no facilities in this analysis that only generate or manage TC wastes.

²¹ State Inspections includes all inspections conducted in the state according to RCRIS, not just those conducted at facilities in this analysis. Gross State Product data were obtained from the Bureau of Economic Analysis.

²² State Violations Last includes all violations found in the state according to RCRIS, not just those found at facilities in this analysis.

on the level of violations last year will not generally be common knowledge to facilities and therefore should not affect the violation decision.²³

Strict Liability is a dummy variable equal to 1 if the state had instituted a strict liability regime by year t .²⁴ Under strict liability, firms are responsible for any damages caused by releases of hazardous waste into the environment regardless of whether the firm acted negligently or with intent. In terms of enforcement behavior, strict liability regimes may give enforcement agencies stronger incentives to find the potentially responsible parties associated with a spill or accident, and thus stronger incentives to inspect facilities, because damages can be recovered from such parties without proof of negligence or intent. Also, because strict liability may impose additional costs on firms, it should create additional incentives for firms to comply with hazardous waste regulations, and thus I expect a negative coefficient on Strict Liability in the Violation equation.

The enforcement burden that a state's environmental agency faces depends on the level of activity in the state. To control for the workload faced by state environmental agencies, many studies use the level of manufacturing or the number of manufacturers in a state. However, because a significant portion of hazardous waste is produced by facilities in sectors that are not considered manufacturing (i.e., service sectors), I use a measure of the state's gross product in millions of dollars in year t .²⁵ A higher workload means less resources are available to inspect the facilities in this analysis. Additionally, states with heavier workloads may be forced to target enforcement efforts and inspections will be less evenly distributed. Thus, I expect the coefficient on Gross Product to be negative in the Inspection equation and positive in the Violation equation.

Environmental Membership measures the proportion of citizens in the state that are members in environmental organizations such as the Sierra Club, which can be seen as a measure of the environmental awareness of the population.²⁶ This variable is included in the Violation equation because Hamilton [6] finds that penalties are higher in states where a larger percentage of citizens are members in environmental organizations. Thus I expect the coefficient on Environmental Membership to be negative in the Violation equation. Environmental Membership is the explanatory variable omitted from the Inspection equation because there is no obvious relationship between citizen membership in environmental organizations and the probability of inspection for a facility. Moreover, when Environmental Membership was included in a probit regression of the Inspection equation, it was not significant.²⁷

Finally, both equations include dummy variables for the Transition years (1990 and 1991) and the Post-Change years (1992 through 1995) to test whether there are

²³ This information is publicly available in RCRIS, although there generally is a lag between violations being detected and entered into RCRIS. Also, EPA's annual report on RCRA Enforcement does not provide state level data.

²⁴ The data on adoption of Strict Liability come from analyses of state Superfund Programs conducted by both the Environmental Protection Agency and the Environmental Law Institute.

²⁵ Gross State Product data was obtained from the Bureau of Economic Analysis, Department of Commerce website (<http://www.bea.doc.gov/bea/regional/gsp/gspsum - c.htm>).

²⁶ Environmental Membership was taken from the 1991–1992 Green Index, Hall and Kerr [5].

²⁷ The probit regression included all of the variables listed in Table II as well as the Environmental Membership variable. Environmental Membership was not significant at the 90% level. Also, there were no changes in the sign or significance of other explanatory variables (as compared to the results presented in Table III) when Environmental Membership was included.

changes in the probability of inspection or the probability of violation due to the change in the penalty policy. These are the key explanatory variables for this study. The trend shown in Table I suggests that the coefficients on these dummies in the Inspection equation should be positive as the number of inspections has increased since 1990. However, it is not obvious whether actual (as opposed to observed) violations have increased or decreased since the policy revision and thus the purpose of this analysis is to determine whether the coefficient on Post-Change in the Violation equation is positive or negative.

5. RESULTS

A. Determinants of an Inspection

The results of the censored bivariate probit are presented in Table III. First, consider the results for the Inspection equation. All of the coefficients in this equation are significant at the 95% level. As expected, the coefficients on Permit, Generated, Managed, Received, High Priority, Medium Priority, Violated Last, and State Inspections are positive. The coefficients on State Violations Last and Gross Product are negative, as expected. However, contrary to expectations, the coefficient on Inspected Last is positive, indicating that firms that were inspected in the last year are more likely to be inspected this year. This suggests that there may be an unobserved or omitted characteristic of the facility that the enforcement agency is targeting. For example, there may be specific activities conducted or substances managed at the facility that make the facility more likely to be

TABLE III
Results of the Censored Bivariate Probit

Variable	Inspection equation		Violation equation	
	Coefficient	Standard error	Coefficient	Standard error
Constant	-0.747*	0.025	-0.861*	0.041
Permit	0.578*	0.021	0.341*	0.020
Generated	0.032*	0.002	0.033*	0.002
Managed	0.003*	0.001	0.000	0.001
Received	0.018*	0.002	0.016*	0.001
High Priority	0.702*	0.018	0.465*	0.019
Medium Priority	0.584*	0.018	0.425*	0.020
Inspected Last	0.585*	0.014	0.007	0.023
Violated Last	0.092*	0.015	0.246*	0.016
TC	0.120*	0.015	0.063*	0.017
State Inspections	0.798*	0.015	0.208*	0.017
State Violations Last	-0.585*	0.022		
Strict Liability	-0.270*	0.012	-0.093*	0.015
Gross Product	-0.273*	0.029	-0.299*	0.037
Environmental Membership			0.014*	0.002
Post-Change	0.141*	0.012	-0.099*	0.015
Transition	0.124*	0.014	0.084*	0.016
ρ	0.868*	0.017		

* Significant at the 95% level.

inspected, or the facility could be conveniently located (close to a regional office or other regulated facilities).

The negative coefficient on Strict Liability is also the opposite of expectations. Rather than increasing the likelihood of inspections, a strict liability regime corresponds to less frequent inspections. One possible explanation is that the additional liability is used as a substitute for monitoring effort. An alternative explanation could be that inspections in states with strict liability are more targeted (that is, less evenly distributed) than in states without strict liability regimes. Because strict liability allows a state to recover damages without proof of negligence or intent, the incentives to target particular types of facilities may be higher.

The coefficient on TC is positive, indicating that facilities that generated or managed TC wastes are more likely to be inspected once the TC wastes are regulated under RCRA, even after controlling for the additional quantities of waste generated and managed at the facility. The coefficients on the dummies for Post-Change and the Transition are both positive and significant as expected, given the increase in the number of inspections over time as shown in Table I. This parallel increase in inspection levels and penalty levels might be due to stronger incentives to monitor facilities and detect violations given the higher penalties. An alternative explanation is that the increase in both inspections and penalties in 1991 was due to an underlying increase in concern over violations, and thus an overall strengthening of EPA's enforcement efforts.

B. Factors Affecting Compliance

For the Violation equation, the majority of coefficients also are significant at the 95% level. The coefficient on Permit is positive, which is consistent with the fact that permitted facilities must comply with a larger set of regulations and thus the costs of compliance are higher for these facilities, as are the potential benefits from violation. Also, this result indicates that the increased incentive to violate the regulations outweighs the increased probability that the facility will be inspected. Similarly, the coefficients on Generated, Managed, and Received are all positive, suggesting that larger facilities have increased incentives to violate regulations (due to higher compliance costs) that also outweigh the higher probability of getting caught. However, while the coefficients on Generated and Received are significant, the coefficient on Managed is not.

The coefficient on Violated Last is positive, which is the opposite of what was expected and in addition is not consistent with the positive coefficient on this variable in the Inspection equation. According to this result, facilities that violated last year are more likely to violate this year, even though penalties are higher for repeat violators and past violators are more likely to be inspected. One possible explanation is that it may take time to correct a violation, particularly if it requires installing new technology, so that past violations may in part determine current violations. Another explanation is that there may be unobserved or omitted variables such as corporate culture that make a facility more or less likely to violate, which this variable is capturing. This could also explain why the coefficients on High Priority and Medium Priority are positive, rather than negative as expected, because high- and medium-priority facilities have had releases to the environment in the past. A second potential explanation for the positive coefficients on High and Medium Priority is that such facilities often have complex waste

management activities, so that the costs of compliance (and thus the benefits from violation) are likely to be high. The coefficient on Inspected Last is also positive, but it is not significant. The coefficient on TC is positive and significant, indicating that facilities with TC wastes were more likely to be in violation after these wastes were regulated, even after controlling for the additional quantities of waste generated and managed at the facility. Facilities may have had to adjust or add waste management systems to accommodate the TC waste streams that could have resulted in additional violations.

Of the state-specific variables in the violation equation, only the coefficient on Strict Liability has the expected sign. The negative coefficient suggests that the additional liability does serve to deter violators by increasing the cost of noncompliance. Also note that the existence of strict liability increases compliance even though it decreases the probability of inspection. The coefficient on State Inspections is positive and significant, rather than negative as expected. However, if the same facilities violate regulations repeatedly, the number of inspections may reflect higher levels of past violations. The coefficient on Gross Product is negative, also contrary to expectations. Additionally, the negative coefficient is inconsistent with the negative coefficient on Gross Product in the Inspection equation. However, states with high gross products are the larger, more industrialized states and thus are likely to have relatively well-established environmental programs that could result in higher compliance levels.

One puzzling result is that Environmental Membership has a positive and significant coefficient, rather than the expected negative coefficient. Thus in states with high levels of environmental membership, facilities are more likely to violate even though evidence suggests that penalties are higher in these states. In this specification of the Violation equation, it appears that Environmental Membership is proxying for other state characteristics because when the model is expanded to include dummies that control for regions, as is discussed later in this section, the coefficient has the expected negative sign.

The coefficient on the dummy for Post-Change is negative and significant, suggesting that the change in the penalty policy did have its intended effect of increasing compliance among the facilities in the analysis. The dummy for the Transition period (1990–1991) is positive and significant, however, indicating that firms did not respond to the policy change immediately, but perhaps waited until states began following the revised penalty schedule. Note also that the correlation coefficient for the two equations, ρ , is positive as expected and statistically significant. This suggests that there may be unobserved or unmeasured characteristics that make facilities more likely to violate the regulations and make a facility more likely to be inspected.

To get a sense of the magnitude of the effect of the revised penalty policy on compliance behavior, I calculated the probability of a violation both before and after the penalty policy at a hypothetical facility that has the mean values for continuous explanatory variables and the median values for discrete explanatory variables. The probability of a violation prior to the revised penalty policy was 23% and dropped to 20% after the revised penalty policy, a decrease of three percentage points (about 15%). At this same hypothetical facility, the probability of inspection was 22% prior to the revised penalty policy and 26% after. In comparison, during the transition period the probability of inspection also rose to 26% but the probability of violation rose to 25% rather than falling. It is interesting to note

that the probability that any facility is actually found in violation averages 24% for the entire period of analysis.

To get a better sense of the size of the effect the other explanatory variables have on both inspections and compliance, Table IV reports the change in the probability (in percentage points) of this hypothetical facility being inspected or violating the regulations that would result from various changes in the explanatory variables. For example, if the facility were a permitted facility, the probability of inspection would increase by 20 percentage points and the probability of violation would increase by 11 percentage points. (The “base” probabilities of inspection and violation are 22 and 23%, respectively.)

For the three quantity variables, I calculated the effect that an increase of one standard deviation would have on the inspection and violation probabilities. These increases have a relatively small effect (3 percentage points or less) on both probabilities. In contrast, for high-priority facilities, the likelihood of inspection increases by 25 percentage points and the likelihood of violation by 16 percentage points. For medium-priority facilities, the probability of inspection increases by only 21 percentage points, but the probability of violation increases by 15 percentage points, which is almost the same as the increase for high-priority facilities. For the most part, changes in the explanatory variables have a larger effect on the probability of inspection than on the probability of a violation. However, having violated last year only increases the probability of inspection by 3 percentage points but increases the probability of violation by 8 percentage points.

One important consideration in evaluating these results is that over the period of analysis there were a number of changes to RCRA regulations. Arguably the most

TABLE IV
Effects of Changes in Explanatory Variables on the Probability of Inspection and Violation at a Hypothetical Facility^a

Change in facility/state characteristics	Change in probability of inspection (in percentage points)	Change in probability of violation (in percentage points)
Not Permitted to Permitted	20	11
Increase Quantity Generated by one std. dev.	3	3
Increase Quantity Managed by one std. dev.	1	<1
Increase Quantity Received by one std. dev.	2	2
Low to High Priority	25	16
Low to Medium Priority	21	15
Not Inspected Last to Inspected Last	21	<1
Not Violated Last to Violated Last	3	8
Not TC to TC	4	2
Increase State Inspections by one std. dev.	11	3
Increase State Violations Last by one std. dev.	-4	NA
Strict Liability to Not Strict Liability	22	3
Increase Gross Product by one std. dev.	< -1	< -1
Increase Env. Membership by one std. dev.	NA	2
From Pre- to Post-Change	5	-3
From Pre-Change to Transition	4	3

^a The basis for comparison is a hypothetical facility that has the mean value for continuous explanatory variables and the median value for discrete explanatory variables.

significant change, the addition of TC wastes to RCRA regulation, has been explicitly included in the analysis. Other changes are not included. Because almost all of the changes to RCRA regulations during the period of analysis acted to increase the level of regulation and the cost of compliance, we might expect the number of violations to be increasing over time relative to inspection levels, *ceteris paribus*. Thus, the results of this analysis can be viewed as a lower bound estimate of the effect of the revised penalty on compliance behavior. Additionally, if there are a significant number of accidental violations that receive relatively small fines (because the revised penalty policy allows the penalty to be adjusted based on "good faith efforts"), this would also bias the results of the analysis towards 0.²⁸

One potential shortcoming of this analysis is the possible existence of unobserved heterogeneity among the facilities in the analysis. The results presented above do suggest that there may be an unmeasured characteristic of facilities that the Agency is targeting or which makes some firms more likely to violate. To explore the possible effects of unobserved heterogeneity on the results I conducted a random effects probit for the Inspection equation alone.²⁹ While the results of this sensitivity analysis suggest that there is unobserved heterogeneity in the data, including random effects into the Inspection equation does not change the sign, magnitude, or significance of any of the coefficients.³⁰ Thus it is not expected that the lack of random effects in the censored bivariate probit model biases the results.

C. Regional Differences

The RCRA penalty policy in part was revised to address differences across EPA regions. Therefore, I also estimated the censored bivariate probit model with regional dummies and regional dummies interacted with the Post-Change dummy to examine differences in inspections and violations across regions both before and after the penalty revision.³¹ As shown in Table V, adding the regional dummies did not change the sign of any of the coefficients for the explanatory variables other than Environmental Membership in the Violation equation. The coefficient on Environmental Membership now has the expected negative sign, which is consistent with Hamilton's [6] finding that penalties are higher in states with higher membership in environmental organizations. Adding the regional dummies also made the coefficient on the Post-Change variable in both equations no longer significant.

While not all of the regional dummies or the regional dummies interacted with the Post-Change dummy are significant, both sets of dummies are jointly significant according to likelihood ratio tests. Also, most coefficients on the regional dummies are positive compared to the omitted region, Region 10 (Alaska, Idaho, Oregon, and Washington), in both the Inspection and the Violation equations, and the majority are significant. Only the coefficient for Region 9 (Arizona, California, Nevada, and Hawaii) in the Inspection equation is negative. The results are more varied for the regional dummies interacted with the Post-Change dummy. For the

²⁸ I thank an anonymous referee for pointing this out.

²⁹ Incorporating random effects into the complete censored bivariate probit model poses computational difficulties.

³⁰ In particular, the null hypothesis that there is no cross-period correlation of error terms across observations for a single facility must be rejected at the 95% confidence level.

³¹ The Appendix provides a list of the 10 EPA regions and the states in each region.

TABLE V
Results of the Censored Bivariate Probit Including EPA Region Dummies

	Inspection equation		Violation equation	
	Coefficient	Standard error	Coefficient	Standard error
Constant	−1.001*	0.045	−0.783*	0.080
Permit	0.595*	0.021	0.333*	0.021
Generated	0.031*	0.002	0.032*	0.002
Managed	0.003*	0.001	0.001	0.001
Received	0.018*	0.001	0.017*	0.001
High Priority	0.724*	0.018	0.469*	0.020
Medium Priority	0.600*	0.018	0.410*	0.020
Inspected Last	0.546*	0.014	0.008	0.023
Violated Last	0.122*	0.016	0.229*	0.016
TC	0.138*	0.015	0.079*	0.017
State Inspections	0.746*	0.019	0.193*	0.019
State Violations Last	−0.672*	0.026		
Strict Liability	−0.240*	0.014	−0.141*	0.019
Gross Product	0.075	0.047	−0.314*	0.061
Environmental Membership			−0.009*	0.004
Post-Change	0.072	0.058	−0.041	0.079
Transition	0.119*	0.014	0.089*	0.016
Region 1	0.254*	0.044	0.380*	0.060
Region 2	0.072	0.046	0.163*	0.060
Region 3	0.220*	0.043	0.225*	0.057
Region 4	0.414*	0.043	0.021	0.061
Region 5	0.255*	0.040	0.224*	0.054
Region 6	0.192*	0.044	0.089	0.063
Region 7	0.268*	0.047	0.474*	0.063
Region 8	0.219*	0.062	0.031	0.074
Region 9	−0.119*	0.051	0.420*	0.073
Region 1 × Post	0.061	0.068	−0.239*	0.091
Region 2 × Post	−0.108	0.066	−0.050	0.088
Region 3 × Post	−0.001	0.065	−0.173*	0.086
Region 4 × Post	0.143*	0.063	0.150	0.083
Region 5 × Post	0.079	0.061	−0.070	0.083
Region 6 × Post	0.040	0.067	−0.096	0.088
Region 7 × Post	−0.203*	0.074	−0.215*	0.098
Region 8 × Post	−0.073	0.097	−0.266*	0.116
Region 9 × Post	0.204*	0.069	−0.062	0.096
ρ	0.860*	0.019		

* Significant at the 95% level.

Inspection equation, only the coefficients for Regions 4, 7, and 9 are significant, with positive coefficients for Regions 4 and 9 and a negative coefficient for Region 7. For the Violation equation, Regions 1, 3, 7, and 8 have negative and significant coefficients. None of the other regional dummies has a significant coefficient.

Using these coefficients, I calculated the probability of inspection and violation at a hypothetical facility in each of the 10 regions both before and after the revised penalty policy, as shown in Table VI. Prior to the policy revision, the probability of a violation ranged from 31% in Region 7 to 16% in Region 10, even though the probability of inspection in Region 7 was higher than that in Region 10. After the policy revision, the probability of violation is much more consistent across regions,

TABLE VI

Effect of Revised Penalty Policy on Compliance Behavior of Hypothetical Facility, by Region

Region	Before revised policy		After revised policy		Difference	
	Prob. of inspection (%)	Prob. of violation (%)	Prob. of inspection (%)	Prob. of violation (%)	Prob. of inspection (%)	Prob. of violation (%)
Region 1	22	27	26	18	4	-8
Region 2	17	20	16	18	-1	-2
Region 3	21	22	23	16	2	-6
Region 4	27	16	34	19	7	3
Region 5	22	22	26	19	5	-3
Region 6	20	18	23	15	3	-3
Region 7	22	31	18	22	-4	-9
Region 8	21	17	21	10	0	-7
Region 9	13	28	19	25	6	-4
Region 10	15	16	17	15	2	-1

with 7 regions in the range 15 to 20%. However, the policy appears to have had a very different effect in some of the regions. For example, in Region 4 violations appear to have increased after the policy revision even though the probability of inspection also increased significantly. In Region 7, on the other hand, violations decreased by nine percentage points even though inspections also decreased by four percentage points. Although not all of the coefficients on the regional dummies are significant, these results suggest that differences in hazardous waste implementation and enforcement across EPA regions persist.

6. CONCLUSIONS

Although a cursory examination of the number of RCRA violations that have been detected over time shows an increase in violations since EPA's Revised RCRA Civil Penalty Policy was published, this analysis shows that the revised policy did have the desired effect of increasing compliance with hazardous waste regulations. The results are consistent with Becker's seminal model of crime and enforcement in which an increase in penalties leads, *ceteris paribus*, to an increase in compliance, although other theoretical work has shown that in some circumstances an increase in penalties will not lead to increased compliance. However, the revised penalty policy increased some penalties 10-fold, while the estimated increase in compliance appears to be on the order of 10 to 20%. Although increasing levels of regulation over the period of this analysis may underestimate the effect of the penalty policy revision, it does not appear that this policy was as successful in increasing compliance as its supporters hoped that it would be.

The findings of this analysis are consistent with a number of other studies that have examined compliance with other environmental regulations. The enforcement strategies of EPA and authorized states target facilities for inspection that are more likely to be violating hazardous waste regulations. Most factors that increase the probability of violation also increase the probability of inspection and thus detection, such as having a RCRA permit or generating large quantities of waste. Also, noncompliance does increase the likelihood that the facility will be inspected

in the future. The large effect that regional dummies have on inspection and violation probabilities, as well as the change in these probabilities due to the revised penalty policies, also suggests that further work should be done to examine differences in the structure and effectiveness of state and regional enforcement programs.

APPENDIX

EPA Regions

Region	States
1	Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont
2	New Jersey, New York, Puerto Rico, ^a Virgin Islands ^a
3	District of Columbia, ^a Delaware, Maryland, Pennsylvania, Virginia, West Virginia
4	Alabama, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee
5	Illinois, Indiana, Michigan, Minnesota, Ohio, Wisconsin
6	Arkansas, Louisiana, New Mexico, Oklahoma, Texas
7	Iowa, Kansas, Missouri, Nebraska
8	Colorado, Montana, North Dakota, South Dakota, Utah, Wyoming
9	Arizona, California, Guam, ^a Hawaii, Nevada
10	Alaska, Idaho, Oregon, Washington

^a In the region, but not included in the analysis due to lack of state data.

REFERENCES

1. G. Becker, Crime and punishment: An economic approach, *J. Polit. Econom.* **76**, 169–217 (1968).

2. P. Bose, Regulatory errors, optimal fines, and the level of compliance, *J. Public Econom.* **56**, 475–484 (1995).

3. W. Gray and M. Deily. Compliance and enforcement: Air pollution regulation in the U.S. steel industry, *J. Environ. Econom. Management* **31**, 96–111 (1996), doi:10.1006/jeem.1996.0034.

4. W. H. Greene, A statistical model for credit scoring, Working Paper No. EC-92-29, Department of Economics, Stern School of Business, New York University (1992).

5. B. Hall and M. L. Kerr, “1991–1992 Green Index: A State-by-State Guide to the Nation’s Environmental Health,” Island Press, Washington, DC (1991).

6. J. T. Hamilton, Going by the (informal) book: The EPA’s use of informal rules in enforcing hazardous waste laws, in “Advances in the Study of Entrepreneurship, Innovation, and Growth,” Vol. 7, pp. 109–155 (1996).

7. E. Helland, The enforcement of pollution control laws: Inspections, violations, and self-reporting, *Rev. Econom. Statist.* **80**, 141–153 (1998).

8. A. Heyes, Cutting environmental penalties to protect the environment, *J. Public Econom.* **60**, 251–165 (1996).

9. A. N. Kleit, M. A. Pierce, and R. C. Hill, Environmental protection, agency motivations, and rent extraction: The regulation of water pollution in Louisiana, *J. Regul. Econom.* **13**, 121–137 (1998).

10. B. Laplante and P. Rilstone, Environmental inspections and emissions of the pulp and paper industry in Quebec, *J. Environ. Econom. Management* **31**, 19–36 (1996), doi:10.1006/jeem.1996.0029.

11. K. K. Lear and J. W. Maxwell, The impact of industry structure and penalty policies on incentives for compliance and regulatory enforcement, *J. Regul. Econom.* **14**, 127–148 (1998).

12. S. F. Leung, How to make the fine fit the corporate crime? An analysis of static and dynamic optimal punishment theories, *J. Public Econom.* **45**, 243–256 (1991).

13. W. A. Magat and W. K. Viscusi, Effectiveness of the EPA’s regulatory enforcements: The case of industrial effluent standards, *J. Law Econom.* **30**, 331–360 (1990).

14. D. J. Poirier, Partial observability in bivariate probit models, *J. Econometrics* **12**, 209–217 (1980).